

MGM1879: TUMOURTRACK - COMPREHENSIVE TUMOR PANEL

Report Details	Specimen Information	Ordering Clinician
Sample ID / Order ID: 9999213 / 1774748	Specimen Site: Brain	Clinician: Dr. Roopesh kumar
Collection Date: 19 th May 2026	Specimen Received: FFPE Tissue Blocks [1]	Affiliation: MGM Healthcare-Chennai
Date Received: 19 th May 2026	Specimen Tested: 1270/26 A	Served By: NA
Report Date & Time: 16 th Jun 2026 11:43 AM	Tumor Content (%): 80	Report Status: Final

Clinical Summary:
Pediatric type diffuse low grade glioma

Test Result Summary					
Potential treatment impacts			Prognostic and Diagnostic findings		Clinical trials
2	0	0	0	0	0
Effective	Ineffective	Safety	Prognostic	Diagnostic	Trials

TMB STATUS
TMB Score: 0.00 mut/Mb



AMP Classification	Evidence Level	Treatment	Treatment Benefit	Drug Approval	Clinical Trials
Genomic variants					
PIK3CA p.Glu545Lys (MISSENSE) Variant Allele Frequency - 5.7%					
Tier IIC	6 Clinical	Alpelisib + Fulvestrant, Capivasertib + Fulvestrant, Inavolisib + Palbociclib + Fulvestrant	Effective	Off-label	0
FUSION variants					
DBN1/BRAF (fusion) Read depth - 205x					
Tier IA	7 Clinically Approved	Tovorafenib	Effective	Approved	0

^ Refer to Glossary section for the classification criteria details.
\$ Drug Approvals are based on US-FDA Guidelines. Kindly refer to local guidelines if required.

Note: Decisions regarding treatment action plan should not be solely based on this test results. These findings are highly recommended to be correlated with the patient’s clinical, pathological, and family history for decisions on diagnosis, prognosis or therapeutics.

There were no other clinically significant variations detected in this subject.

ADDITIONAL BIOMARKERS DETECTED (Variant of Uncertain Significance)

This section provides information about variants that do not have any therapeutic value. However, these variants may or may not have a likely oncogenic effect.

No other biomarker that warrants to be reported was detected

GLOSSARY

AMP Classification Criteria: Displays the classification of a biomarker according to the recommendations of the Association for Molecular Pathology (AMP) [PMID: 27993330].

Tier	Criteria
Tier IA	Variants of strong clinical significance. FDA-approved therapy or biomarkers included in professional guidelines.
Tier IB	Variants of strong clinical significance. Well-powered studies with consensus from experts in the field.
Tier IIC	Variants of potential clinical significance. FDA-approved therapies for different cancer types or investigational therapies. Multiple small published studies with some consensus.
Tier IID	Variants of potential clinical significance. Preclinical trials or a few case reports without consensus.
Tier III	Variants of unknown clinical significance.
Tier IV	Benign or likely benign variants.

Evidence Level: The Evidence Level, values 7-1 indicate the reliability of a biomarker to predict a specific patient outcome. This can include predictive treatment effects; in this case, the scores 7-1 apply for biomarkers associated with a single drug or drug combination. The Evidence levels are defined as follows:

Score	Definition
7, Clinically approved	Variant approved by a regulatory agency to predict a specific effect (i.e., response, resistance, or toxicity) in the patient's disease or cancer type.
6, Clinical	- Variant has not yet been approved by a regulatory agency for the patient's disease but has been observed in at least one large cohort study to predict a specific effect of the drug (i.e., to be effective, resistance) in the patient's disease. - The biomarker has been approved by a regulatory agency to predict a specific effect of the drug (response, resistance) with other diseases.
5, Clinical	- Variant has not yet been approved by a regulatory agency for the patient's disease but has been observed to predict a specific effect of the drug (i.e., response, resistance) on patients with other diseases or conditions. - For variants predicting a drug to be effective or resistant, there is evidence from some patients in several cohort studies and preclinical evidence.
4, Clinical	- The variant has not yet been approved by a regulatory agency for the patient's disease. However, this variant has been observed to predict a specific effect of the drug (i.e., response, resistance) on patients with other diseases or conditions. - For variants predicting a drug to be effective or resistant, there is evidence from a few clinical case reports and additional preclinical evidence. - For variants predicting a drug to be toxic, there is evidence from a prospective study, >1 retrospective studies, or >1 cohort studies.
3, Preclinical	Variant has not yet been observed/tested in patients to predict a specific effect but has been observed in preclinical experiments. There is experimental evidence from cell lines or mouse models, for example.
2, Preclinical	Variant has not yet been observed/tested in patients or preclinical models to predict a specific effect. However, this effect can be inferred when drug-sensitivity data are available for another variant and if the two variants have the identical functional impact on the same downstream pathway

Score	Definition
1, Preclinical	Variant has not yet been observed/tested in patients or preclinical models to predict a specific effect. However, this effect can be inferred when drug-sensitivity data are available for another variant and if both variants have the identical functional impact on the protein.

Drug approval:

The development stage of the treatment for the patient's indication as per US-FDA guidelines.

Stage	Definition
Approved	This drug is launched for the primary or a secondary patient disease
Off-Label	This drug is launched for a disease other than the primary or secondary patient diseases
Investigational	This drug is currently under clinical development in the patient disease.
Other	None of the other stages are applicable. The drug or drug class is, for example, suspended, discontinued, or withdrawn.

Medications with potential for adverse reaction or ineffectiveness: Refers to the identified treatments that are predicted to be associated with negative physiological responses to a drug therapy (i.e., drug resistance and toxicity) based on the biomedical evidence.

Potential impact: The specific drug effect predicted by the identified mutation (i.e. response, resistance, or toxicity).

Treatment: The generic name of the therapeutic agent listed on the report.

ACTIONABLE BIOMARKER DETAILS

DBN1/BRAF

Gene Fusion: *DBN1:BRAF*

Total Read Depth: 205x

Exon/Intron: E:12

5'Gene *DBN1*

Ensemble Gene ID: ENST00000393565.6

Exon/Intron: I:10

5'Chromosome Breakpoint (hg38):

chr5:177459145:-

3'Gene: *BRAF*

Ensemble Gene ID: ENST00000646891.2

3'Chromosome Breakpoint (hg38):

chr7:140781706:-

Gene Summary 1: The protein encoded by *DBN1* is a cytoplasmic actin-binding protein thought to play a role in the process of neuronal growth. It is a member of the drebrin family of proteins that are developmentally regulated in the brain. A decrease in the amount of this protein in the brain has been implicated as a possible contributing factor in the pathogenesis of memory disturbance in Alzheimer's disease. At least two alternative splice variants encoding different protein isoforms have been described for *DBN1*.

Gene Summary 2: *BRAF* encodes a protein belonging to the RAF family of serine/threonine protein kinases. This protein plays a role in regulating the MAP kinase/ERK signaling pathway, which affects cell division, differentiation, and secretion. Mutations in *BRAF*, most commonly the V600E mutation, are the most frequently identified cancer-causing mutations in melanoma, and have been identified in various other cancers as well, including non-Hodgkin lymphoma, colorectal cancer, thyroid carcinoma, non-small cell lung carcinoma, hairy cell leukemia and adenocarcinoma of lung. Mutations in *BRAF* are also associated with cardiofaciocutaneous, Noonan, and Costello syndromes, which exhibit overlapping phenotypes. A pseudogene of *BRAF* has been identified on the X chromosome.

Clinical and Therapeutic Relevance: The serine/threonine-protein kinase BRAF activates the RAS/MAPK signaling pathway to promote cell proliferation and survival. *BRAF* fusion proteins likely lead to constitutive *BRAF* kinase activity that activates downstream signaling. **The type II RAF inhibitor tovorafenib is indicated for patients with relapsed or refractory pediatric low-grade glioma (LGG) harboring a *BRAF* rearrangement.** In the phase 2 clinical trial FIREFLY-1 (n=77), tovorafenib showed an overall response rate (ORR) of 69% (10 complete responses (CR), 31 partial responses (PR)) for *BRAF*-fusion-positive pediatric LGG patients, the majority of whom harbored the *KIAA1549-BRAF* fusion.

PubMed References: [37978284](#)

PIK3CA(p.Glu545Lys)

Gene: *PIK3CA*

Nucleotide change: chr3:g.179218303G>A

cDNA change: c.1633G>A

Exon: 10

Protein change: p.Glu545Lys

Variant Type: MISSENSE

Variant Allele Frequency: 5.7%

Population MAF: 0 (1000G); 0 (gnomAD);

In-silico Predictions: D(SIFT); D(LRT);

PrD(Polyphen2)

Transcript ID : ENST00000263967.4

Variant Allele Depth: 8x

Gene Function: Oncogene

Gene Summary: Phosphatidylinositol 3-kinase is composed of an 85 kDa regulatory subunit and a 110 kDa catalytic subunit. The protein encoded by *PIK3CA* represents the catalytic subunit, which uses ATP to phosphorylate PtdIns, PtdIns4P and PtdIns(4,5)P2. *PIK3CA* has been found to be oncogenic and has been implicated in cervical cancers. A pseudogene of *PIK3CA* has been defined on chromosome 22.

PIK3CA(p.Glu545Lys)

Clinical and Therapeutic Relevance: *PIK3CA* is the catalytic subunit of the lipid phosphoinositide-3-kinase (PI3K) that activates the PI3K/AKT signaling pathway to promote cell proliferation and survival. This variant strongly activates the downstream pathway in cell lines and lies outside hotspot regions. The AKT inhibitor capivasertib in combination with fulvestrant is indicated for adult patients with hormone receptor (HR)-positive, HER2-negative locally advanced or metastatic breast cancer (BC) with this variant. The PI3K inhibitor alpelisib in combination with fulvestrant is indicated as second-line therapy in HR-positive/HER2-negative locally advanced or metastatic BC with *PIK3CA* mutations (EMA) or with hotspot mutations in exon 8, 10, or 21 (ESMO guidelines). Inavolisib, in combination with palbociclib and fulvestrant, is indicated for adults with endocrine-resistant, *PIK3CA*-mutated, HR-positive, HER2-negative, locally advanced, or metastatic BC. A retrospective study found that thirteen patients (8.3%) had *PIK3CA* mutations, which were linked to younger age (49.4 vs. 58.1 years, $p = 0.02$). The mutation was associated with shorter PFS (6.9 vs. 12.4 months, $p = 0.01$) and OS (21.2 vs. 24.2 months, $p = 0.049$). It also correlated with more disseminated disease at diagnosis (46.2% vs. 11.1%, $p = 0.004$). Overall, *PIK3CA* activating mutations were associated with earlier recurrence and reduced survival in adult glioblastoma [PMID: 31036078].

PubMed References: 40454641, 39901702, 39236276, 38837894, 37916956, 37256976, 35759724, 35671774, 35133871, 31934607, 31678634, 31632573, 31091374, 28183140, 27177864, 26469692, 26325104, 25426553, 22422409, 21362200, 19633047, 31036078

DISCLAIMER

- Decisions regarding treatment action plan should not be solely based on these test results. These findings are highly recommended to be correlated with the patient's clinical, pathological, radiological and family history for decisions on diagnosis, prognosis, or therapeutics.
- The therapy information provided in this report is based on FDA approved drugs data, NCCN guidelines, peer-reviewed published literature, standard clinical databases, and strength of biomarker results. These therapies may or may not be suitable/beneficial to a particular patient. This clinical report summarizes potentially effective medications, potentially ineffective medications, and medications that may pose a higher risk of adverse reactions by mapping the patient's genetic alterations to the biomedical reference information. The report may also provide prognostic and diagnostic biomarkers detected or shown for the given disease context.
- The clinical trials information provided in this report is compiled from www.clinicaltrials.gov as per currently available data, however completeness of information provided herein cannot be guaranteed. This information should only be used as a guide and specific eligibility criteria should be reviewed thoroughly for the concerned patient. MedGenome Labs does not guarantee or promise an enrolment in any clinical trials.
- The identification of a genomic biomarker does not necessarily imply pharmacological effectiveness or ineffectiveness. The medications identified by the treating physician may or may not be suitable for use on a particular patient. Thus, the clinical report does not guarantee that any particular agent will be effective in the treatment of any particular condition. Also, the absence of a treatment option does not determine the effectiveness or predict an ineffective or safety-relevant effect of a medication selected by the treating physician.
- The classification and clinically relevant information for the reported variants is based on peer-reviewed publications, public clinical databases, medical guidelines (NCCN, ASCO, AMP) or other publicly available information and it has been ensured that the information provided is up to date at the time of report generated, however continuous updates may happen in public domains. Also, the classification of variants can change based on the updated literature evidence. Re-analysis of the results can be requested at additional cost.
- This test is performed on the patient's tumor sample without a paired blood sample; therefore, it may include variations which may be of germline origin. However, this test is designed and validated for the detection and reporting of somatic genomic variants only and does not discriminate between germline and somatic variants. If clinically warranted, appropriate germline testing and genetic counselling for the patient should be considered for further evaluation.
- Due to poor quality of FFPE tissue blocks, the QC parameters for extracted DNA and RNA may not pass to proceed further with the testing, therefore there is a possibility of assay failure at various steps (DNA/RNA QC, Library QC, Bioinformatics QC) or compromised results that include low gene coverage and low variant depth. However, sample status in such scenarios shall be sent through mail to the ordering clinician.
- This test has been validated at MedGenome Labs and the limit of detection (LOD) of allele fraction for SNVs and short InDels is $\geq 5\%$ and for fusions is ≥ 10 spanning reads. However, the report may include, at the discretion of laboratory director, the variants with lower allele burden (3-5%) having strong or potential clinical significance or those have been reported earlier in the patient. Variants with $< 1\%$ allele fraction and variants of uncertain significance with $< 5\%$ allele fraction are not routinely reported. However, possibility of false negative or false positive below the limit of detection of this assay cannot be ruled out.
- The sensitivity for the detection of Copy Number Variations (CNVs)/Gene amplifications is $> 90\%$ in this assay. Confirmatory tests such as FISH are recommended to confirm the same.
- Variations occurring in non-coding and deep intronic regions are not detected in this assay.
- TMB Score is dependent on the number of the genes in the panel and role of cytosine deamination artifacts (C>T) due to poor tissue fixation cannot be ruled out. Our threshold for tissue deamination is $< 10\%$.

- TMB testing is an evolving biomarker and there are no universal guidelines for the number and choice of genes and methodology used for the calculation of TMB.
- Different cancers and gene panels have different cut-offs for Low, Intermediate and High TMB status. The current validation on this panel is based on a limited subset of diverse cancer types, such as breast cancer, colon cancer, lung cancer, oral cancer, skin cancer, stomach cancer, and testicular cancer in 25 samples and reference standard cell lines with known TMB value.
- **Additional case specific disclaimer : None**

TEST DESCRIPTION

The MedGenome's Comprehensive Tumor panel is a high throughput next-generation sequencing based single assay to detect cancer causing genomic alterations such as base substitutions, insertions, deletions in the coding exons (+/- 10 bp flanking intronic region), copy number variations, fusions and tumor mutation burden (TMB) in the patient's tumor that may provide treatment benefit to the patient. This multigene tumor agnostic panel covers key cancer genes for the assessment of various biomarkers in solid tumor malignancies as per NCCN guidelines. All known and unknown partners for the fusion genes covered in this panel are reported. The genes related to Homologous recombination repair (HRR) pathway are included in this test.

TEST METHODOLOGY

Sample type: FFPE (Formalin-Fixed Paraffin-Embedded) Specimen; A histopathologic review is performed to determine the tumor content in the FFPE block/curls.

Extraction and Library Preparation: Tumor nucleic acid is extracted from the FFPE tissue block and used to perform UMI-based target enrichment and sequencing using a custom capture kit.

Sequencing: QC-passed libraries are sequenced on a validated Illumina sequencing platform to a minimum depth >2000X (pre-UMI) and compressed to >250X (post-UMI) for variant analysis.

Data Analysis: The sequences obtained are aligned to the human reference genome (GRCh38/hg38) using the BWA program [PMID: [19451168](#), PMID: [23155063](#)] and the STAR aligner [PMID: [23104886](#)]. Somatic mutations are identified using the UMI corrected Sention pipeline [PMID: [31481971](#)], designed to accurately detect all classes of genomic alterations.

Fusion Analysis: The sequenced reads are pre-processed and aligned to the Human reference genome/ transcriptome model (GRCh38/hg38) using the STAR aligner [PMID: [23104886](#)]. Gene fusions are determined using multiple published fusion detection programs that have been validated and fine-tuned to our assays [<https://www.biorxiv.org/content/10.1101/011650v1>, <https://www.biorxiv.org/content/10.1101/120295v1>, PMID: [21593131](#)]. The fusions were prioritised based on the total read depth (number of spanning reads and split reads) [PMID: [26019724](#)].

Reporting: The variants are annotated using our in-house annotation pipeline. Reportable genomic alterations and fusions are prioritized, classified, and reported based on AMP-ASCO-CAP guidelines [PMID: [27993330](#)] and NCCN guidelines.

Limit of Detection (LOD): The LOD for SNVs and InDels is 5% Variant Allele Frequency (VAF), fusions is >10X read depth and for CNVs is > 6 copy numbers.

Tumor mutation burden (TMB): Tumor mutation burden is a hybrid capture-based custom assay that involves single primer extension of target regions barcoded with unique molecular indexes for tumor profiling and provides an accurate assessment of mutation load (mutations/Mb). The assay detects and annotates low-frequency somatic variants (SNPs) from 231 genes related to tumor and immune biology, encompassing a total of 1.06Mb of the genomic DNA. The TMB score is calculated as the number of non-synonymous somatic variants across the panel length and normalized to Whole Exome Sequencing (WES), and calculated as mutations per megabase of the genome. The TMB status is considered low if the TMB score is lower than 10; intermediate if the TMB score is between 10 and 15; and high if the TMB score is larger than 15. The Threshold is chosen based on internal benchmark analyses of lung cancer cell lines and different tissue cancer samples [PMID: [31471895](#)].

The transcript used for clinical reporting generally represents the canonical transcript (according to Ensembl release 109 human gene model), which is usually the longest coding transcript with strong/multiple supporting evidence. However, clinically relevant variants annotated in alternate complete coding transcripts could also be reported. Variants annotated on incomplete and nonsense-mediated decay transcripts are not reported.

§This test is developed, and its performance characteristics is determined by MedGenome Labs Ltd.

GENES ANALYSED

SNVs, InDels, CNVs and TMB									
ABRAXAS1	AKT1	AKT2	AKT3	ALK	APC	AR	ARID1A	ARID1B	ARID2
ATM	ATR	ATRX	B2M	BAP1	BARD1	BCL2	BCL6	BCORL1	BIRC3
BLM	BRAF	BRCA1	BRCA2	BRIP1	BTK	CARD11	CCND1	CCND2	CCND3
CCNE1	CD28	CD58	CD79A	CD79B	CDH1	CDK12	CDK4	CDK6	CDKN2A
CDKN2B	CEBPA	CHD8	CHEK1	CHEK2	CIC	CIITA	CREBBP	CSF1R	CSF3R
CTNNB1	CXCR4	DDR2	DDX3X	DICER1	DNMT3A	EGFR	EIF1AX	EP300	ERBB2
ERBB3	ERBB4	ERCC1	ERCC2	ERCC3	ERCC4	ERCC5	ERG	ESR1	EZH2
FANCA	FANCB	FANCC	FANCE	FANCI	FANCL	FBXW7	FGF1	FGF10	FGF14
FGF19	FGF2	FGF23	FGF3	FGF4	FGF5	FGF6	FGF7	FGF8	FGF9
FGFR1	FGFR2	FGFR3	FGFR4	FLT1	FLT3	FOXL2	FOXO1	GEN1	GNA11
GNA13	GNAQ	GNAS	H3-3A	H3C2	HNF1A	HRAS	ID3	IDH1	IDH2
INPP4B	IRF4	ITPKB	JAK2	JAK3	KDM6A	KDR	KEAP1	KIT	KLF4
KMT2A	KMT2C	KMT2D	KRAS	LAMP1	MAP2K1	MAP2K2	MCL1	MDM2	MDM4
MEF2B	MET	MFHAS1	MLH1	MLLT3	MN1	MPL	MRE11	MSH2	MSH3
MSH6	MTOR	MUTYH	MYB	MYBL1	MYC	MYCL	MYCN	MYD88	NBN
NF1	NF2	NFE2	NOTCH1	NOTCH2	NOTCH3	NPM1	NRAS	NRG1	NTRK1
NTRK3	PALB2	PDGFRA	PDGFRB	PIK3CA	PIK3CB	PIK3CD	PIK3CG	PIK3R1	PIM1
PMS2	POLD1	POLE	PPARG	PPP2R2A	PRDM1	PTCH1	PTEN	PTPN11	RAD50
RAD51	RAD51B	RAD51C	RAD51D	RAD54L	RAF1	RB1	RET	RHOA	RICTOR
ROS1	RPS6KB1	SDHA	SDHAF2	SDHB	SDHC	SDHD	SF3B1	SLX4	SMAD4
SMARCA4	SMARCB1	SMARCE1	SMO	SOCS1	SOX11	SRC	STAT3	STAT5B	STAT6
STK11	SUFU	TCF3	TERT	TET2	TFRC	TG	TGFB2	TNFAIP3	TNFRSF14
TP53	TRAF3	TRAF7	TSC1	TSC2	TSHR	VHL	XPC	XPO1	XRCC1
XRCC2	ZAP70								

Copy number variations are analyzed in all the genes listed above. However, the number values will be provided for the genes highlighted in Blue.

Fusions									
ABL1	AKT3	ALK	AR	ASPL	AXL	BCL2	BCL6	BCOR	BRAF
BRCA1	BRCA2	CCND1	CCND2	CDK4	CDKN2A	COL1A1	CSF1	CSF1R	CTLA4
DNAJB1	DUSP22	EGFR	EML4	EPC1	ERBB2	ERG	ESR1	ETS1	ETV1
ETV4	ETV5	ETV6	EWSR1	FARSB	FGFR1	FGFR2	FGFR3	FGFR4	FLI1
FLT1	FLT3	FLT4	FOS	FOSB	FUS	GLI	GNA14	GRB7	HEY1
HMGA2	HOXA1	HOXA10	HOXA11	HOXA13	HOXA2	HOXA3	HOXA4	HOXA5	HOXA6
HOXA7	HOXA9	INI1	IRF4	ITK	JAK2	JAZF1	KDR	KIF5B	KIT
KMT2A	LCK	LMO1	LMO2	LYL1	MAML2	MDM2	MEAF6	MEN1	MET
MGEA5	MKL2	MLLT3	MSH2	MYB	MYBL1	MYC	MYOD1	NAB2	NCOA1
NCOA2	NCOA3	NOTCH1	NOTCH2	NOTCH3	NRG1	NTRK1	NTRK2	NTRK3	NUTM1
OGA	PAX3	PAX7	PAX8	PDGFB	PDGFRA	PDGFRB	PHF1	PI3K	PICALM
PIK3CA	PIK3R2	PLAG1	PLCG1	PPARG	PRKACA	PRKCA	PRKCB	PRKCD	PTH
PTPBR	RAF1	RELA	RET	RNF213	ROS1	RPS5KB1	RPS6KB1	RSPO2	RSPO3
SAS	SLC5A5	SS18	SSX1	SSX2	SUZ12	SYN2	TAF15	TAF2N	TCF12

TCF3	TCL1A	TFE3	TFEB	TFG	THADA	TLX1	TLX3	TMPRSS2	TP53
TP63	TPM3	UACA	USP6	VGLL2	VGLL3	WWTR1	YAP1	YWHAE	ZC3H7B

CLINICAL TRIALS

The following trials are potentially best suited for your patient's indication, considering all reported treatment recommendations. See <https://clinicaltrials.gov> (clinical trials from NCT) or <https://trialsearch.who.int> (clinical trials from other registries) for more information.

Note: The Phase 1 and Phase 2 clinical trials will be provided on request.

Clinical trials in total : 0 Trial countries : IN-India, US-United States

S.No	Title	Phase and ID	Intervention	Disease	Age & Sex
No Clinical Trials.					

Himadri S. Baul

Dr Syed Muqlisur Rehman

Himadri Shekhaar Baul, M.Sc
Lead Genome Analyst
Solid Tumor Oncology

Dr Syed Muqlisur Rehman, MD (Path)
Molecular Pathologist
KMC Registration No. 71468

END OF REPORT